Vector spaces Revision

Friday, May 22, 2026 10:10 AM

A set $\vec{V}$ contain certain objects; if these objects follow certain (10) properties. This set is a vector space.

$+$, "x" scalar multiplication — usual operations

$u+v, \in \vec{V}$

$u+v = (u_1+v_1, u_2+v_2)$

$k\vec{u} = (ku_1, ku_2)$

1. $u+v \in V$ ∴ closure
2. $ku \in V$ closed under scalar multiplication
3. $u+v = v+u$ Commutative
4. $u+(v+w) = (u+v)+w$ Associative
5. $k(u+v) = ku+kv$ distributive "x" over addition
6. $(k+m)\vec{u} = k\vec{u} + m\vec{u}$ " " + " x
7. $k(m\vec{u}) = (km)\vec{u}$ Associative "x"
8. $\vec{u}+0 = 0+\vec{u} = u$ Additive identity
9. $\vec{u}+(-u) = \vec{0}$ Additive inverse
10. $1\vec{u} = \vec{u}\cdot 1 = \vec{u}$ multiplicative identity.

$5-5 \Rightarrow 5+(-5)$

$\frac{7}{2} \Rightarrow 7\times\frac{1}{2}$

$5\times\frac{1}{5} = 1$

$\vec{V} = \{\text{Polynomial's set}\}$

$0 \Rightarrow 0x^0$

$\vec{u}$, $\vec{v}$

$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_2x^2 + a_1x + a_0$

$n=0$, $a_0 \Rightarrow$ Constant Polynomial

$n=1$; $a_1x+a_0 \Rightarrow$ Linear

$n=2$ $a_2x^2+a_1x+a_0$ quadratic.

1. $u+v \Rightarrow$ Polynomial

   $u+v \in \vec{V}$ — I
2. $k\vec{u} \in \vec{v}$ — II
3. $\vec{u}+\vec{v} = \vec{v}+\vec{u}$ — III
4. $u+(v+w) = (u+v)+w$ — IV
5. $u+0 = \vec{u}$ — V + identity
6. $u+(-u) = 0$ — VI + inverse

$u = x^2+3x+5$, $-u = -x^2-3x-5$

$v = 2x-4$

$w = 5$

$k=3$, $m=4$

$7x^2+21x+35$

(6) $u + (-u) = 0$ — (vi) + inverse

(7) $(K+m)\vec{u} = K\vec{u} + m\vec{u}$ — (vii) Distributive + over ×

(8) $K(\vec{u}+\vec{v}) = K\vec{u} + K\vec{v}$ — (viii) Distributive × over +

(9) $(Km)\vec{u} = K(m\vec{u})$

(10) $\vec{u} \cdot 1 = \vec{u}$

$7x^2+21x+35$

$3x^2+9x+15 + 4x^2+12x+20$

$7x^2+21x+35$

$3(x^2+5x+1)$

$= 3x^2+15x+3$

$6x-12$

$3x^2+9x+15$

$3(4x^2+12x+20)$

$\boxed{12x^2+36x+60}$

Q ⇒ Prove / disprove $\vec{V} = \{$ Set of Polynomials having degree $\leq 5\}$

$-u = -x^3-4x^2-5x-6$

$\vec{U} = x^3+4x^2+5x+6$

$\vec{V} = 4x^2+5x+6$

$\vec{W} = 5x+6$

$U = p_5(t)$

(1) closed '+' Scalar multiplication

(2) $5\vec{U} =$

(3), 4, (5) $\vec{U}+\vec{0} = \vec{U}$

(6) $u+(-u)=0$ ⇒ + inverse

(7) $K(U+V) = 2\vec{U} + 2\vec{V}$

(8) $(K+m)u = (K+m)p_5t$

$= Kp_5t + mpt$

(9) $K(mU) = (Km)U$

(10) $1 \cdot \bar{u} = u$

Prove / disprove $\vec{V}: \{$ Polynomials degree $\geq 5\}$

$U = x^6 - 5x$ $V = -x^6 + 6x$

$U = x^6 - 5x$ $\qquad V = -x^6 + 6x$

$U+V = 0 + x = x^1 \in \vec{V}$

"0" doesn't Exist

Not a vector Space

1. Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2), \quad k\mathbf{u} = (0, ku_2)$$

(ku_1, ku_2)

(a) Compute $\mathbf{u} + \mathbf{v}$ and $k\mathbf{u}$ for $\mathbf{u} = (-1, 2)$, $\mathbf{v} = (3, 4)$ and $k = 3$.

$3\vec{u} = (0, 6)$

(b) In words, explain why V is closed under addition and scalar multiplication.

(c) Since addition on V is the standard addition operation on R^2, certain vector space axioms hold for V because they are known to hold for R^2. Which axioms are they?

$u+v = v+u$

$u+(v+w) = (u+v)+w$

(d) Show that Axioms 7, 8, and 9 hold.

(e) Show that Axiom 10 fails and hence that V is not a vector space under the given operations.

$\vec{u} = (-1, 2)$, $\quad -\vec{u} = -1 \times \vec{u}$

$= -1(-1, 2)$

$\vec{u} + (-u) = (-1, 2) + (0, -2) = (0, -2)$

$= (-1, 0) \neq 0$

$ku + kv$

$K(\vec{u} + \vec{v}) = K[u_1 + v_1, u_2 + v_2]$

$= [0, K(u_2 + v_2)]$

$= (0, Ku_2) + (0, Kv_2)$

$= K(u_1, u_2) + K(v_1, v_2)$

$= Ku + Kv$

Axiom 10 $\quad 1\vec{u} = \vec{u} \qquad 1(u_1, u_2) = (0, u_2) \neq u$

2. Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1 + 1, u_2 + v_2 + 1), \quad k\mathbf{u} = (ku_1, ku_2)$$

$\longrightarrow \in V$ $\quad u+v,\ ku \in V$

(a) Compute $\mathbf{u} + \mathbf{v}$ and ku for $u = (0, 4)$, $v = (1, -3)$, and $k = 2$.

(b) Show that $(0, 0) \neq \mathbf{0}$.

(c) Show that $(-1, -1) = \mathbf{0}$.

(d) Show that Axiom 5 holds by producing an ordered pair $-\mathbf{u}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ for $\mathbf{u} = (u_1, u_2)$.

(e) Find two vector space axioms that fail to hold.

$(-1,-1)$

$(1,1) + (0,0) \neq (2,2)$

$0 \neq 0$

$u = (u_1, u_2)$

$(u_1, u_2) + (-1,-1) = (u_1 \cancel{-1} \cancel{+1}, u_2 \cancel{-1} \cancel{+1})$

$= (u_1, u_2) = \vec{u}$

$\vec{u} + I = \vec{u}$

$(-1,-1) = 0$ Additive

$(u_1, u_2) + (-1,-1) = (u_1 \cancel{-1} \cancel{+1}, u_2 - 1 + 1)$

$= (u_1, u_2)$

$= u$

Find additive identity of $\vec{u}$.

$u = (u_1, u_2)$ $\qquad -u = (-u_1, -u_2)$

$u + (-u) = (\cancel{u_1} - \cancel{u_1} + 1, \cancel{u_2} - \cancel{u_2} + 1)$

$= (1,1)$

$-u$? $\qquad u + (-u) = 0$ Additive.

$-u = (a, b)$ $\qquad = (u_1, u_2) + (a, b)$

$= (u_1 + a + 1, u_2 + b + 1) = (-1, -1)$

$u_1 + a + 1 = -1, \qquad u_2 + b + 1 = -1$

$a = -u_1 - 2, \qquad b = -u_2 - 2$

$-u = (-u_1 - 2, -u_2 - 2)$

$-u = (-u_1 - 2, \quad -u_2 - 2)$

$(u_1, u_2) + (-u_1 - 2, \ -u_2 - 2) = (\cancel{u_1} - \cancel{u_1} - 2 + 1, \quad \cancel{u_2} - \cancel{u_2} - 2 + 1)$

$= (-1, -1)$

$\vec{u} = (u_1, u_2) \Rightarrow$ u+ve

$\boxed{= (u_1 + v_1 - 2, \quad u_2 + v_2 - 2)}$ ✓

What is additive Identity

$\vec{U} + \vec{A} = \vec{U} \Rightarrow (u_1, u_2) + (a, b) = (u_1, u_2)$

$(u_1 + a - 2, \quad u_2 + b - 2) = (u_1, u_2)$

$\cancel{u_1} + a - 2 = \cancel{u_1}, \quad \cancel{u_2} + b - 2 = \cancel{u_2}$

$(2, 2) \qquad a = 2, \quad b = 2$

Find additive Inverse of $\vec{U} = (u_1, u_2)$

$-\vec{u} = (c, d)$

$\vec{u} + (-u) = (2, 2)$

$(u_1, u_2) + (c, d) = (2, 2)$

$(u_1 + c - 2, \quad u_2 + d - 2) = (2, 2)$

$u_1 + c - \cancel{2} = \cancel{2}, \qquad u_2 + d - 2 = 2$

$c = 4 - u_1, \qquad d = 4 - u_2$

$-u = (4 - u_1, \ 4 - u_2) \qquad U = (u_1, u_2)$

$= (\cancel{u_1} + 4 - \cancel{u_1} - 2, \quad \cancel{u_2} + 4 - \cancel{u_2} - 2)$

$(2, \quad 2)$

Additive Identity is Not Zero All times.

It depends upon operation define

Simil $(-u_1, -u_2)$ is not Additive Inverse all The times